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Ministry of Education

Jouf University



College of Computer and Information Sciences

THESIS

In fulfillment of the requirements for the degree of
Bachelor in Computer and Network Engineering

Follow Me Robot

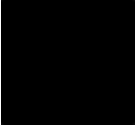
Assistant Robot that Follows Human

PROJECT SUPERVISOR

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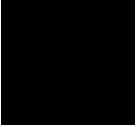
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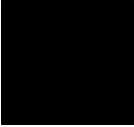
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DECLARATION

We hereby declare that this work, in its entirety or part, has not been copied from any source. It is also stated that we created this software and that accompanying report is entirely the result of our efforts. If any part of this project is found to be copied from another source or reproduction of another. We will stand firm in the face of the consequences. No part of the work presented here has been submitted in support of any other degree or qualification offered by this or any other university or institute of learning.



CERTIFICATION

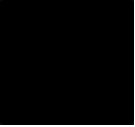
It is certified that the contents and form of the project entitled "Follow Me Robot-Assistant Robot that Follows Human" submitted by
. Have been satisfactory for the requirements for the award of the degree of Bachelor's
in computer and network engineering.

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ABSTRACT

Human following robot is very common technology and it's come a long way and has been revolutionary in assisting humans to complete tasks. The communication between the robot and the human is the most critical component to act independently. Our main goal is to design and build a robot that is not just for tracking humans, it's also to provide assistance. To make the project we used RFID Technology, a person who the robot must follow hold a RFID tag. The robot will recognize it and start follow's it.

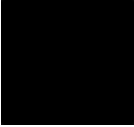


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INTRODUCTION

1.1 Goals

In the modern era characterized by rapid technological advancements, robotics has become an essential aspect of human life. The objective of this thesis is to devise an autonomous robotic system that is capable of processing RFID tags and localizing them. The passive UHF (Ultra High Frequency) RFID technology was employed for tracking the target. The robot was designed to traverse obstacles and maintain a safe distance while following the designated individual autonomously.

1.2 Motivation

The first challenge of an autonomous robot is how to handle a highly unstructured and dynamic environment. To perform the task, sensors must be used to perceive the environment surrounding the robot. On the other hand, the localization of passive RFID tags is another challenge for the RFID community. Investigating the design, implementation, and challenges of human-following robots using RFID technology allows you to tackle real-world problems and devise potential solutions that can positively impact society. The process of researching, writing, and communicating on this topic helped us to develop critical thinking, communication, and technical skills

1.3 Introduction

In the past few years, robotic technology has evolved substantially. To aid humans in their work, a robot must be able to recognize and follow them, necessitating the development of robots like the "Follow Me Robot-Assistant Robot that Follows Human" , that can interact and co-exist with them. Localizing the items in the surroundings as well as the robot itself is a significant problem in enabling the robot for executing a range of activities in the real-world. A robot needs to be sophisticated to track a human

through traffic regions, a vibrant atmosphere, and both indoors and out. The project's goal in building the following robot is for it to be able to assist alongside humans. It can be utilized in agriculture, construction, airports, and shopping malls. RFID-enabled human-following robots combine the unique identification capabilities of RFID technology with the versatility of autonomous robotic systems. By utilizing RFID tags to identify and track individuals, these robots can maintain a safe distance while following their designated targets, thus offering valuable assistance.

1.4 Project Scope

The project scope outlines the boundaries, objectives, deliverables, and constraints of the RFID-based human-following robot system. Here's an overview of the project scope:

1. Objective:

- a. Develop a robotic system capable of autonomously tracking and following a human operator using RFID technology.
- b. Enhance human-robot interaction and facilitate various applications such as assistance in warehouses, guiding visitors in public spaces, or aiding individuals with mobility impairments.

2. Deliverables:

- a. A functional prototype of the RFID-based human-following robot system.
- b. Documentation including system architecture, hardware specifications, software design.
- c. Demonstration videos showcasing the robot's capabilities and performance in real-world scenarios.

3. Boundaries:

- a. The project will focus on developing the core functionality of the human-following robot system, including RFID tag detection, human tracking, and obstacle avoidance.
- b. Advanced features such as machine learning algorithms, multi-robot collaboration, and integration with smart environments may be

considered for future iterations but are not within the scope of the current project.

4. Constraints:

- a. Budget constraints: The project budget will dictate the selection of components, materials, and development resources.
- b. Time constraints: The project timeline will be limited, and milestones must be achieved within the allocated time frame.
- c. Technical constraints: The system must operate within the limitations of the selected hardware and software platforms, considering factors such as processing power, memory, and sensor capabilities.

5. Assumptions:

- a. It is assumed that the RFID technology will provide reliable detection and tracking of human operators in various environments.
- b. The system will be developed using off-the-shelf components and open-source software libraries, leveraging existing resources and expertise within the project team.

6. Risk Management:

- a. Risks such as technical challenges, resource constraints, and changes in project requirements will be identified, assessed, and mitigated throughout the project lifecycle.

7. Success Criteria:

- a. The project will be considered successful if the developed prototype demonstrates reliable RFID tag detection, accurate human tracking, effective obstacle avoidance, and intuitive user interaction.
- b. Success will also be measured by the system's performance in real-world scenarios and its potential for further development and commercialization.

By defining the project scope, resources, and efforts toward achieving the defined objectives within the established constraints and boundaries. Regular monitoring and

evaluation will ensure that the project remains on track and delivers the intended outcomes.

1.5 Problem Description

The problem that the RFID-based human-following robot system aims to address is the need for enhanced human-robot interaction and assistance in various environments. In many settings, there is a growing demand for robots that can autonomously navigate and assist humans in tasks such as guiding visitors, providing support in warehouses, or aiding individuals with mobility impairments. However, traditional methods of human-robot interaction, such as manual control or pre-programmed paths, may be limited in their flexibility and responsiveness to dynamic environments and user needs.

Specifically, the problem can be described as

follows:

1. **Limited Autonomy:** Existing robotic systems may lack the ability to autonomously track and follow humans in real-time, requiring constant manual control or predefined routes.
2. **Poor Adaptability:** Robots may struggle to adapt to changes in their environment or the behavior of human operators, leading to inefficiencies and safety concerns.
3. **Inadequate Interaction:** Traditional human-robot interaction methods may be cumbersome or unintuitive, hindering effective communication and collaboration between humans and robots.
4. **Safety Concerns:** Without robust obstacle detection and avoidance capabilities, robots may pose safety risks to both humans and themselves, especially in crowded or dynamic environments.
5. **Limited Accessibility:** Individuals with mobility impairments may face barriers to accessing assistance or navigating public spaces independently,

highlighting the need for innovative solutions to improve accessibility and inclusivity.

By addressing these challenges, the RFID-based human-following robot system aims to provide a reliable, adaptive, and user-friendly solution for enhancing human-robot interaction, improving assistance capabilities, and promoting autonomy and accessibility in various domains.

1.6 Proposed Solution

The proposed solution is to develop an RFID-based human-following robot system that leverages RFID technology to autonomously track and follow a human operator in real-time. This system will address the limitations of traditional human-robot interaction methods by providing a responsive, adaptive, and user-friendly solution for various applications. Here's an overview of the proposed solution:

1. RFID Tag Detection:

- a. The robot will be equipped with an RFID reader module capable of detecting RFID tags carried by the human operator.
- b. Upon detecting an RFID tag, the robot will identify the unique tag identifier to distinguish the human operator from other individuals in the environment.

2. Human Tracking:

- a. Once the RFID tag is detected, the robot will initiate tracking and follow the human operator at a safe distance.
- b. It will adjust its speed and direction in real-time to maintain a constant distance from the human operator while avoiding obstacles in its path.

3. Obstacle Avoidance:

- a. The robot will utilize onboard sensors, such as ultrasonic or LiDAR sensors, to detect obstacles in its path.

- b. It will autonomously navigate around obstacles while continuing to follow the human operator, ensuring safe and efficient movement in dynamic environments.

4. Power Management:

- a. The robot will incorporate efficient power management mechanisms to optimize battery life during operation.
- b. It may support automatic recharging or battery replacement for prolonged usage, ensuring uninterrupted service in various environments.

5. Safety Features:

- a. Safety mechanisms, such as emergency stop buttons and collision detection sensors, will be implemented to prevent accidents or injuries during operation.
- b. The robot's speed and proximity to the human operator will be configurable to ensure safe interaction in diverse environments.

6. Scalability and Adaptability:

- a. The system architecture will be designed to be scalable and adaptable, allowing for future enhancements and integration with additional sensors or features.
- b. It will support customization and configuration to meet the specific requirements of different applications and environments.

7. Testing and Validation:

- a. The proposed solution will undergo rigorous testing and validation procedures to ensure functionality, reliability, and safety in real-world scenarios.
- b. User feedback and usability testing will be conducted to assess the system's effectiveness and user satisfaction, driving continuous improvement and refinement.

Overall, the proposed RFID-based human-following robot system offers a comprehensive solution for enhancing human-robot interaction, improving assistance capabilities, and promoting autonomy and accessibility across various domains.

1.7 Conclusion

In conclusion, the RFID-based human-following robot system constitutes a significant breakthrough in the realm of robotics and human-robot interaction. This system employs RFID technology to autonomously track and follow human operators in real-time, offering a novel solution for various applications. Its ability to navigate dynamic environments, avoid obstacles, and maintain a safe distance from the human operator demonstrates its potential for warehouse logistics and assisted living support, among other areas. The objective of this project is to overcome the limitations of traditional human-robot interaction methods and advance the development of more intuitive, responsive, and user-friendly robotic systems. This technology has the potential to enhance accessibility, improve assistance capabilities, and promote autonomy, thereby embodying the convergence of technology and human-centric design, driving innovation in robotics, and shaping the future of human-robot collaboration.

SYSTEM ANALYSIS AND HARDWARE COMPONENT

2.1 Introduction

The purpose of this chapter is to provide analysis of the proposed system's requirements and constraints to make sure it follows the human effectively and efficiently in the real world. The system analysis will talk about the system's requirements and specific requirements and functional and non-functional needs, and how well it works.

2.2 Requirement

The successful development and implementation of the human-following robot using RFID technology require the fulfillment of several essential requirements. These requirements encompass various aspects of the system, including hardware, software, performance, safety, and ethical considerations. Here is an outline of the primary requirements for this project:

1. Hardware Requirements:

- RFID reader with sufficient range and accuracy for tag detection, Ultrasonic sensor for precise distance measurement
- Servo motors and motor drivers for navigation
- Batteries and power management system for extended operation
- Communication module for receiving commands and transmitting data

2. Software Requirements:

- Efficient control algorithms for navigation, obstacle avoidance, and following behavior
- Integration of RFID detection, distance measurement, and navigation subsystems

3. Performance Requirements:

- Reliable and accurate RFID tag detection, Smooth and stable following behavior, Adaptability to changing environments and conditions
- Energy efficiency for prolonged operation
- Safe and efficient obstacle avoidance

4. Safety Requirements:

- Safe interaction with humans and the environment
- Reliable emergency stop and fail-safe mechanisms
- Thorough testing and validation of system performance and safety

By addressing these requirements, the project aims to develop a human-following robot using RFID technology that is safe, and efficient with the potential to enhance safety, and efficiency across various applications.

2.3 Specific Requirement

1. Sensors:

- RFID Reader Module:
 - Frequency: UHF (Ultra-High Frequency) for longer range and better penetration.
 - Detection Range: Adjustable, typically up to several meters for optimal human tracking.
 - Antenna Type: Directional or omnidirectional depending on the application requirements.
- Obstacle Detection Sensors:

- Type: Ultrasonic, LiDAR, or infrared sensors for detecting obstacles in the robot's path.
- Coverage: 360-degree coverage to ensure comprehensive obstacle detection.
- Range: Adjustable range for detecting obstacles at various distances.
- Motors:
 - Resolution: High resolution for precise motion control.

2. Power:

- Battery:
 - Type: Lithium-ion (Li-ion) battery for high energy density and long runtime.
 - Capacity: Sufficient capacity to support continuous operation for several hours.
 - Voltage: Compatible with the robot's motor controllers and electronics.

3. Communication:

- Wireless Communication:
 - Protocol: Wi-Fi or Bluetooth for real-time data exchange between the robot and the human operator.
 - Range: Sufficient range to maintain communication in large indoor or outdoor environments.
- Data Transmission:
 - Rate: High data transfer rate for transmitting sensor data, control commands, and status updates.
 - Reliability: Error correction and packet retransmission mechanisms to ensure reliable communication.

4. Mechanics:

- Chassis:
 - Material: Lightweight yet durable material such as aluminum or carbon fiber.
 - Design: Compact and streamlined design for maneuverability in crowded spaces.
- Wheels:
 - Type: Omni-directional or differential drive wheels for omnidirectional movement.
 - Size: Appropriate wheel diameter and width for stability and traction on various surfaces.
- Payload Capacity:
 - Sufficient payload capacity to carry additional equipment or payloads as needed.

5. Environment:

- Indoor and Outdoor Operation:
 - Compatibility: Designed to operate seamlessly in both indoor and outdoor environments.
 - Weatherproofing: Weather-resistant components and sealing to protect against moisture and dust ingress.
 - Temperature Range: Operating temperature range suitable for different climates and weather conditions.

6. Safety:

- Collision Avoidance:

- Collision Detection Sensors: Front-facing sensors to detect obstacles and stop or maneuver the robot to avoid collisions.
- Emergency Stop:
 - Redundant emergency stop buttons for immediate halting of robot motion in case of emergencies.
- Speed and Proximity Control:
 - Adjustable speed and proximity settings to ensure safe interaction with the human operator.

7. Performance:

- Tracking Accuracy:
 - High tracking accuracy to maintain constant distance from the human operator.
 - Real-time Adjustment: Responsive control algorithms for real-time adjustment of speed and direction.
- Motion Smoothness:
 - Smooth acceleration and deceleration profiles for comfortable human tracking without abrupt movements.
 - Stability: Low center of gravity and stable motion characteristics to prevent tipping or instability.
- Battery Life:
 - Optimized power management algorithms for maximizing battery life without sacrificing performance.

8. Estimate Cost:

- Cost-Effective Components:
 - Selection of cost-effective sensors, actuators, and electronic components without compromising quality and performance.
 - The estimated cost of the project is 400 SAR.

- Scalability:
 - Scalable system architecture to accommodate future upgrades or enhancements as needed without significant additional cost.

These specifications provide a comprehensive overview of the key requirements for each component of the RFID-based human-following robot system, ensuring functionality, reliability, safety, and performance while optimizing cost-effectiveness and scalability.

2.4 Functional Requirement

Functional requirements define the specific behavior and functionality of a system. The functional requirements for the RFID-based human-following robot system:

1. RFID Tag Detection:
 - The robot shall be capable of detecting RFID tags carried by the human operator within a predefined range.
 - The RFID reader module shall identify unique tag identifiers to distinguish between multiple individuals in crowded environments.
2. Human Tracking:
 - Upon detecting the RFID tag, the robot shall initiate tracking and follow the human operator at a safe distance.
 - The robot shall adjust its speed and direction in real-time to maintain a constant distance from the human operator while avoiding obstacles.
3. Obstacle Avoidance:
 - The robot shall utilize onboard sensors (e.g., ultrasonic or LiDAR) to detect obstacles in its path.
 - It shall autonomously navigate around obstacles while maintaining proximity to the human operator.

4. Power Management:

- The robot shall incorporate efficient power management mechanisms to optimize battery life during operation.
- It shall support automatic recharging or battery replacement for prolonged usage.

These functional requirements ensure that the system performs essential tasks such as detecting RFID tags, tracking humans, avoiding obstacles, and managing power effectively.

2.5 Non-Functional Requirement

Non-functional requirements define the qualities or attributes that characterize how the system behaves rather than what it does. Here are non-functional requirements for the RFID-based human-following robot system:

1. Reliability:

- a. The system shall operate reliably under various environmental conditions, including indoor and outdoor settings with different lighting and noise levels.
- b. It shall minimize the occurrence of false detections and unintended movements.

2. Performance:

- a. The robot shall demonstrate responsive tracking behavior with minimal latency between RFID tag detection and robot movement.
- b. It shall achieve smooth and stable motion while following the human operator, ensuring a comfortable user experience.

3. Scalability:

- a. The system architecture shall be scalable to accommodate additional features or sensors for future enhancements.

- b. It shall support integration with other robotic systems or IoT devices for extended functionality.

4. Safety:

- a. Safety mechanisms, such as emergency stop buttons or collision detection sensors, shall be implemented to prevent accidents or injuries during operation.
- b. The robot's speed and proximity to the human operator shall be configurable to ensure safe interaction in dynamic environments.

5. Maintainability:

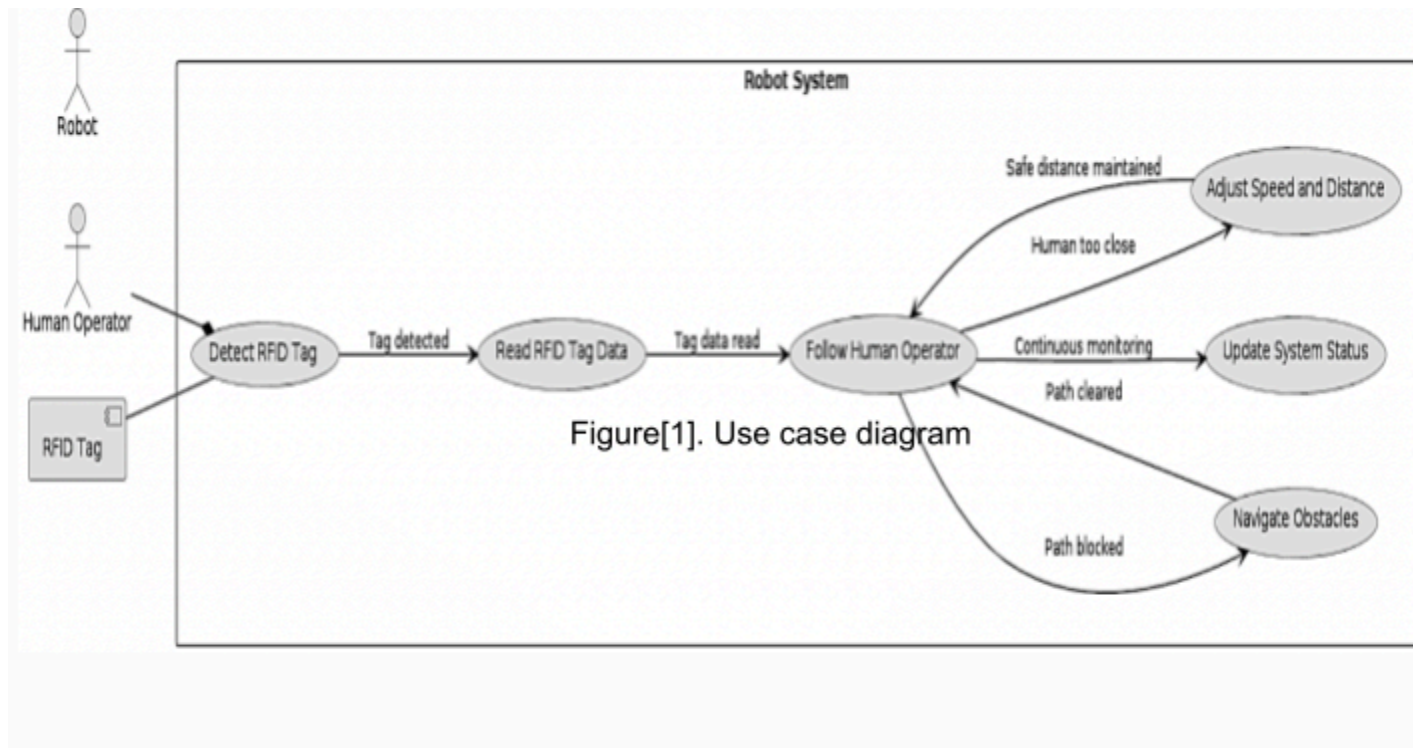
- a. The system shall be designed for ease of maintenance, with modular components that can be replaced or upgraded as needed.
- b. Technical support and documentation shall be provided to assist users in troubleshooting issues and performing routine maintenance tasks.

6. Environmental Adaptability:

- a. The system shall be adaptable to different indoor and outdoor environments, including areas with varying floor surfaces, lighting conditions, and crowd densities.
- b. It shall withstand environmental factors such as temperature fluctuations, humidity, and exposure to dust or moisture.

These non-functional requirements ensure that the system exhibits desirable qualities such as reliability, performance, scalability, usability, safety, security, maintainability, and environmental adaptability, enhancing its overall effectiveness and user satisfaction.

2.6 Use case diagram



Figure[1]. Use case diagram

2.7 Arduino Uno



Figure[2]. Arduino Uno 4 Wi-Fi

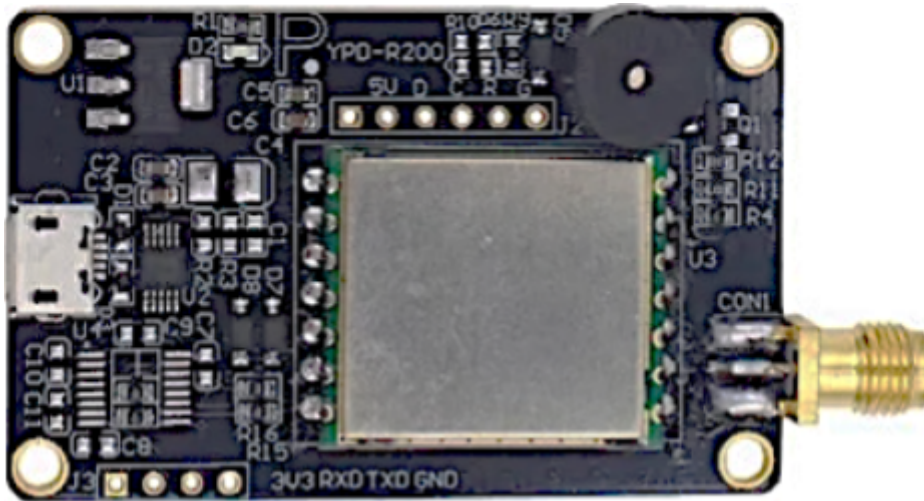
Figure[2]. Arduino Uno 4 Wi-Fi

The Arduino UNO R4 WiFi is an advanced microcontroller board based on the Renesas RA4M1 32-bit microprocessor, which operates at 48 MHz. It is an upgraded version of the classic Arduino UNO board, offering improved speed, memory, and connectivity options. One notable feature of the Arduino UNO R4 WiFi is its ESP32 module, which enables Wi-Fi and Bluetooth 4.2 (BLE) connectivity. This makes it an ideal platform for various IoT applications such as robotics, home automation, and sensor networks.

The board has a USB Type-C connector for both programming and power supply. It features 14 digital input/output pins (6 of which support PWM output), 6 analog input pins, and a 2x15-pin expansion header with various interfaces such as GPIO, I2C, SPI, UART, and analog pins. The board's operating voltage is 3.3V, and its input voltage ranges from 7V to 12V. The Arduino UNO R4 WiFi is equipped with 2 MB of flash memory for program storage, 128 KB of SRAM for data storage, and 2 KB of EEPROM for storing configuration data or other permanent information. This versatile and powerful board is compatible with the Arduino Integrated Development Environment

(IDE) and can be programmed using the popular Arduino programming language. Its wide operating temperature range of -40°C to 85°C makes it suitable for various environments and applications.

2.8 RFID Reader (IN-R200)



Figure[3]. RFID Reader IN-R200

The IN-R200 is an ultra-high frequency (UHF) RFID reader module designed for mid-range applications. Built around the Impinj R2000 RFID engine, this compact reader provides high performance and flexibility, making it ideal for various use cases such as asset tracking, inventory management, and access control. Supporting ISO18000-6C (EPC C1G2) and ISO18000-6B standards, the IN-R200 is compatible with a wide range of RFID tags and systems. It features up to four antenna ports, allowing for better coverage and more flexible installation options. Communication interfaces include USB, RS232, RS485, and Ethernet, enabling easy integration with various devices and systems. With a read range of up to 2 meters, depending on the transponder and environmental conditions, the IN-R200 offers reliable detection for mid-range applications. The reader module consumes low power, operating at 7W, allowing for energy-efficient operation. Additionally, its compatibility with various

development platforms such as Arduino, Android, and Raspberry Pi makes the IN-R200 suitable for embedded solutions. As a versatile and cost-effective solution, the IN-R200 RFID Reader is an excellent choice for applications requiring accurate and reliable RFID detection within a mid-range distance. Its compact size and flexible configuration options make it easy to integrate into existing systems or serve as the foundation for new RFID-based projects.

2.9 RFID Tags

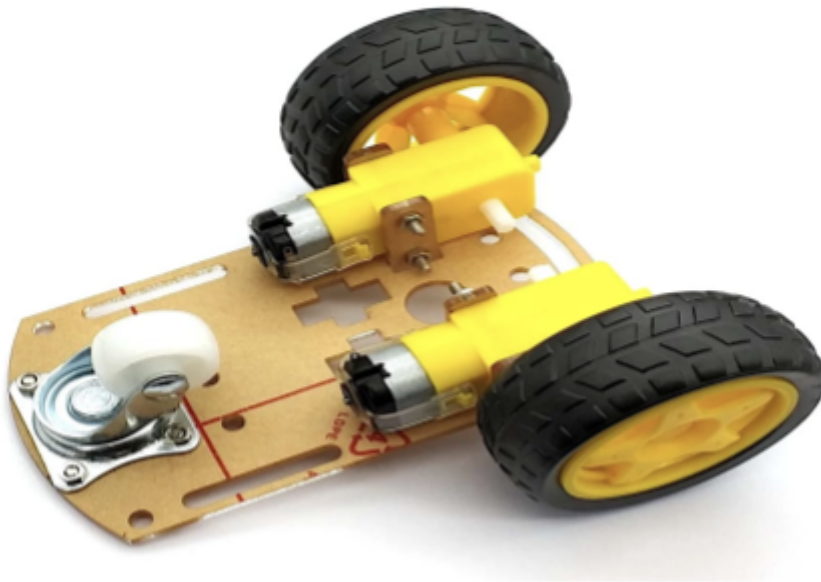


Figure[4]. UHF RFID U8 tag

The UHF RFID U8 tag is a high-performance, passive RFID tag designed for various applications, including logistics management, supply chain tracking, asset management, and inventory control. Operating within the ultra-high frequency (UHF) band, typically between 860 MHz to 960 MHz, it complies with ISO/IEC 18000-6C and EPC Global Class1 Gen2 standards. As a passive RFID tag, the U8 does not require an internal power source; instead, it draws energy from the RF signal transmitted by the RFID reader. It boasts a long read range of up to 8 meters (26 feet) depending on the environment and reader configuration, enabling efficient and accurate tracking of tagged items. With a high storage capacity of 128 bits to 512 bits for user memory, the U8 tag can store essential information such as identification numbers, serial numbers, or

other custom data. Its durable construction ensures resistance against water, dust, and common chemicals, making it suitable for use in various industrial and outdoor environments. Advanced security features, such as data encryption and password protection, are supported by the U8 tag to ensure data integrity and prevent unauthorized access to stored information. The UHF RFID U8 tag is an ideal choice for applications that require reliable and efficient tracking of assets or inventory items over extended distances. Its compatibility with standard RFID systems facilitates seamless integration into existing infrastructures.

2.10 Two wheel drive Robot Chassis with Motors

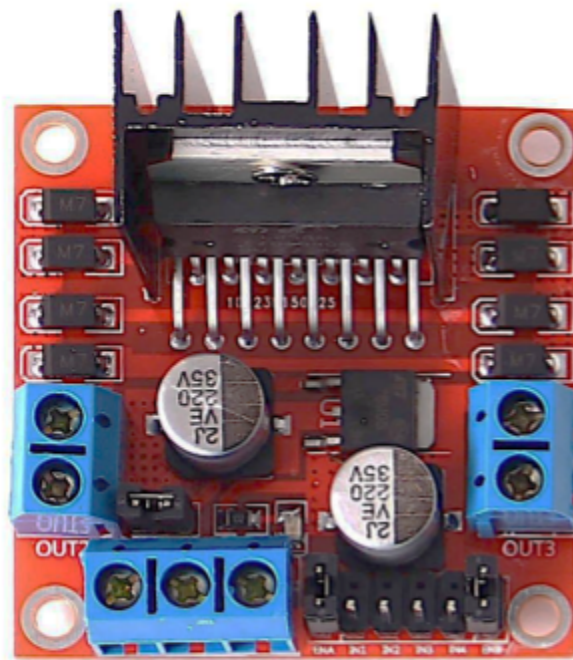


Figure[5]. Two wheel drive Robot Chassis with Motors

A Two-Wheel Drive (2WD) Robot Chassis with Motors is a simple and efficient robotic platform used as a base for building mobile robots or small vehicles. It consists of a chassis or frame, two motors, and wheels that provide propulsion

and steering for the robot. The modular and versatile design of the 2WD Robot Chassis allows for easy customization and compatibility with popular microcontrollers, making it suitable for various robotics projects and educational activities. These kits are user-friendly, with easy assembly and minimal soldering required, catering to both beginners and experienced robot builders.

2.11 Motor Driver(L298N)



Figure[6]. Motor Driver(L298N)

The L298N is a widely-used motor driver module based on the L298 dual H-bridge motor driver integrated circuit (IC). It enables you to control the speed and direction of two DC motors or one stepper motor, handling motor voltages between 5V and 35V with a peak current of up to 2A per channel. Featuring a dual H-bridge design, the L298N Motor Driver is efficient in motor control and offers control inputs for motor

direction and speed. It is equipped with status LEDs for motor power and direction indication, along with an onboard 5V regulator that can be enabled or disabled as needed. The L298N Motor Driver is popular in robotics and hobby projects due to its ease of use, compatibility with microcontrollers such as Arduino, and the ability to control multiple motors simultaneously. By interfacing the motor driver with a microcontroller and providing the appropriate control signals, users can develop various motor-based applications like robotic vehicles and automated systems.

2.12 UltraSonic Sensor (HC-SR04)



Figure[7]. UltraSonic Sensor (HC-SR04)

The HC-SR04 is an ultrasonic sensor commonly used in electronic projects for distance measurement and object detection. It operates by emitting high-frequency sound waves and measuring the time it takes for the sound to bounce back after hitting an object. This allows the sensor to determine the distance between itself and the object based on the speed of sound. The HC-SR04 sensor comprises two main components: a transmitter (or "trigger") and a receiver. The transmitter emits an ultrasonic pulse, and the receiver listens for the

echo. The sensor has four pins: VCC, GND, Trigger, and Echo. The VCC and GND pins are connected to the power supply, while the Trigger and Echo pins are connected to the microcontroller for initiating distance measurements and receiving the results.

Key specifications of the HC-SR04 include:

- Operating voltage: 5V
- Detection range: 2cm to 400cm
- Accuracy: 3mm
- Ultrasonic frequency: 40 kHz

The HC-SR04 is compatible with popular microcontrollers like Arduino, making it a popular choice for a variety of projects involving distance measurement, obstacle avoidance, and object detection.

2.13 Conclusion

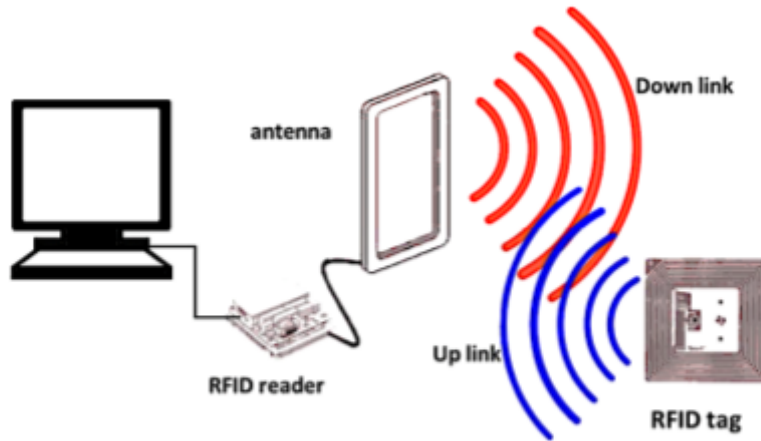
In conclusion, the human-following robot using RFID technology relies on the efficient integration and optimization of these subsystems. By analyzing each subsystem, we can identify potential improvements and optimizations that will contribute to the overall performance, adaptability, and safety of the robot.

HARDWARE OVERVIEW

3.1 History of RFID

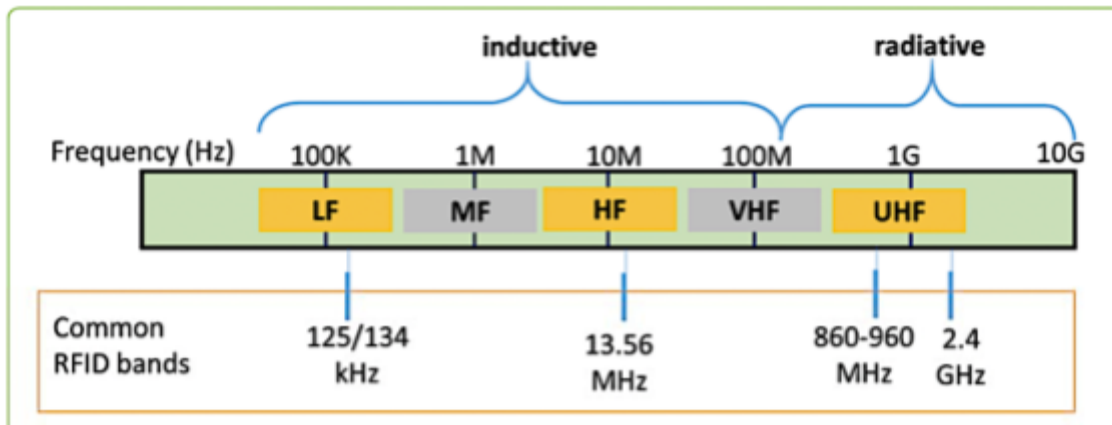
The basic idea of RFID (Radio Frequency Identification) technology, communication using reflected radio energy, can be traced back to the origin of radar technology. The British developed the first active system to identify friend or foe (IFF). A transmitter was deployed on each British plane, when it received the signal from the radar, it replied with a signal that allowed for the identification of the plane. The modern RFID system works the same as this basic concept, when a signal is sent to a RFID tag, it wakes up and either reflects back a signal. Today, 13.56 MHz RFID systems are widely used for access control, payment systems and contactless smart cards. The automobile industry introduced the 13.56 MHz RFID system as an anti-theft device in cars. A reader in the steering column of a car reads the passive RFID tag in the key. The reader will check the ID number feedback from the key tag, and compare it with the programmed ID, if they do not match the car will not start. In the early 1990s, an ultra-high frequency (UHF) RFID system was developed by IBM, it offered up to 20 feet read range and higher data transfer rate. Later, this technology was sold to Intermec, a bar code systems provider. Then, Intermec's RFID systems were deployed in warehouse tracking, farming. But the high cost of RFID systems prevented the widespread deployment of RFID technology.

3.2 Architecture of a RFID System and Type of RFID



Figure[8]. Typical RFID system

A typical RFID system consists of an interrogator, more often called a reader, transponders, also called tags, and the antennae that are connected to the reader for sending and receiving signals. Figure[8] shows a typical RFID system. The antennae may be integrated with the reader, like a hand-held system or separate from the reader and connected by a cable. The tag is integrated with the antenna and at least one silicon chip, which responds to the reader query and stores data to built-in memory. In most cases, the reader will be connected to a computer to provide interface with other systems or human beings.



Figure[9]. Typical frequency spectrum available for RFID

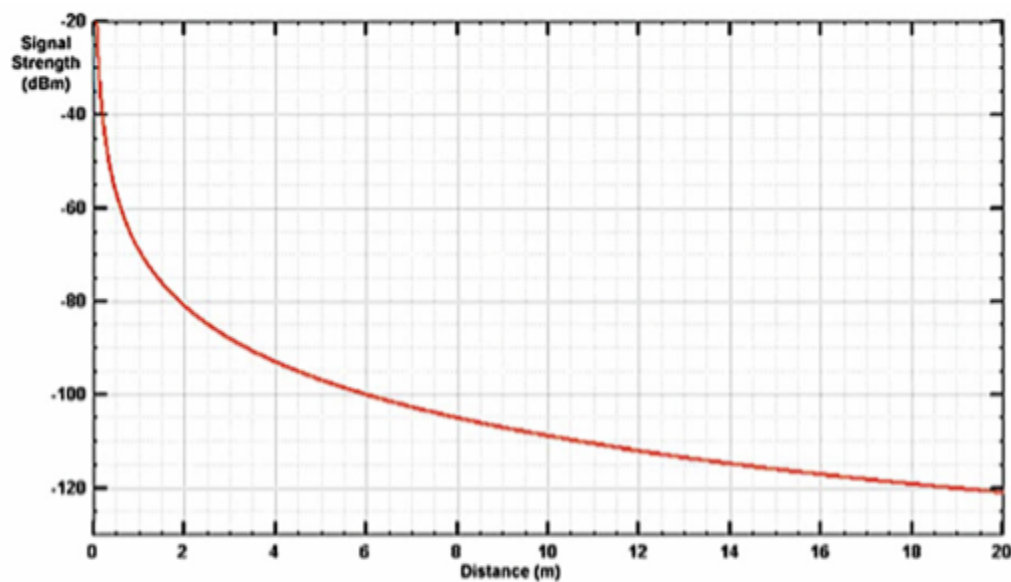
RFID systems can use frequencies from around 100 kHz to over 5 GHz, the RFID frequency bands are shown in Figure [9]. The frequency of most RFID systems are concentrated in very narrow bands that are available by regulators for unlicensed industrial implementation. The most commonly used frequency bands are 125/134 kHz, 13.56 MHz, 860-960 MHz and 2.4 GHz. The RFID systems that rely on 125/134 kHz radio waves are called low frequency (LF) systems. While systems that work on 13.56 MHz are referred to as high frequency (HF) systems. Systems that operate in 860-960 MHz are called ultra-high-frequency (UHF) systems, and the systems in 2.4 GHz are referred to as microwave systems. Nowadays, most retail deployments of UHF RFID systems are in their warehouses and sales floors environments.

3.3 Basics of Passive RFID

Without a built-in battery the passive RFID tag extracts power from the electromagnetic field that is created by the reader. The extracted energy is used to power up internal circuits. There are two different power extracting techniques or coupling techniques, inductively coupling and radiative coupling, used by passive tags.

3.4 Characteristic of UHF Passive RFID

we are interested in is the detectable range of a UHF passive RFID reader. A passive RFID tag without any built-in power supply, absorbs power from a radio wave that is sent by the reader, hence the power intensity of the radio wave determines the detectable range of a UHF passive RFID system.



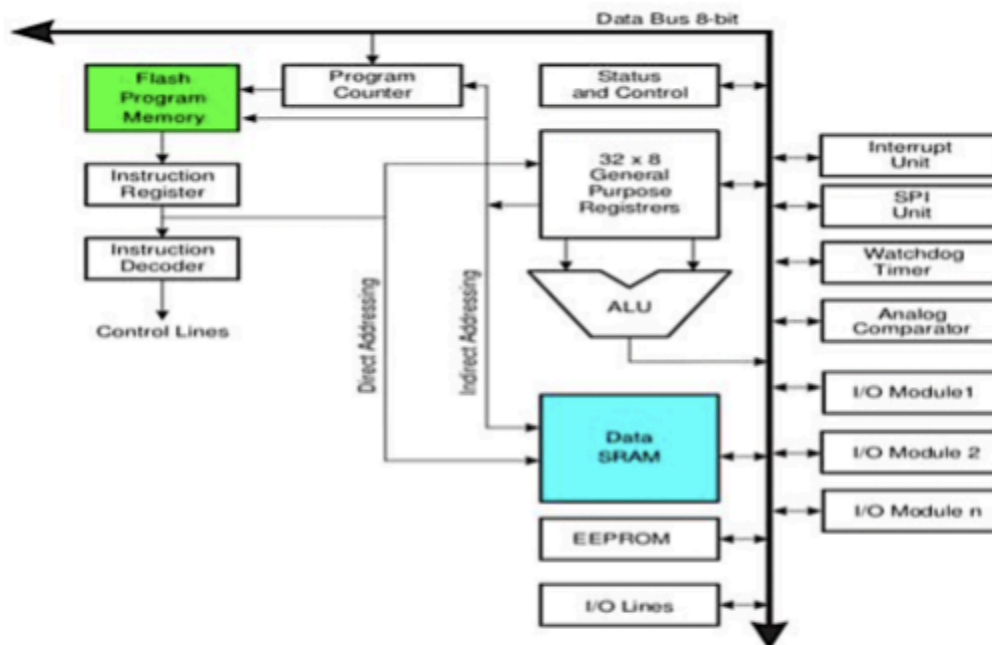
Figure[10]. Typical RFID signal strength over distance

From the above figure[3], we can see that as the distance between RFID tag and RFID antenna, the received power by tag decreases, the probability of detecting that tag decreases tremendously. So, when our robot does need to move as close as it can to read the tag more accurately.

3.5 Introduction of Arduino

Microcontrollers are integrated circuit devices that contain a microprocessor, peripherals, and inputs and outputs in a small physical package. Microcontrollers are at the heart of embedded computing systems widely used in agricultural applications ranging from greenhouses to field robots. The Arduino UNO is a programmable, open-source microcontroller widely used to teach microcontroller principles and programming. The Arduino UNO is a complete hardware and software package that can be used to teach both novice and advanced students.

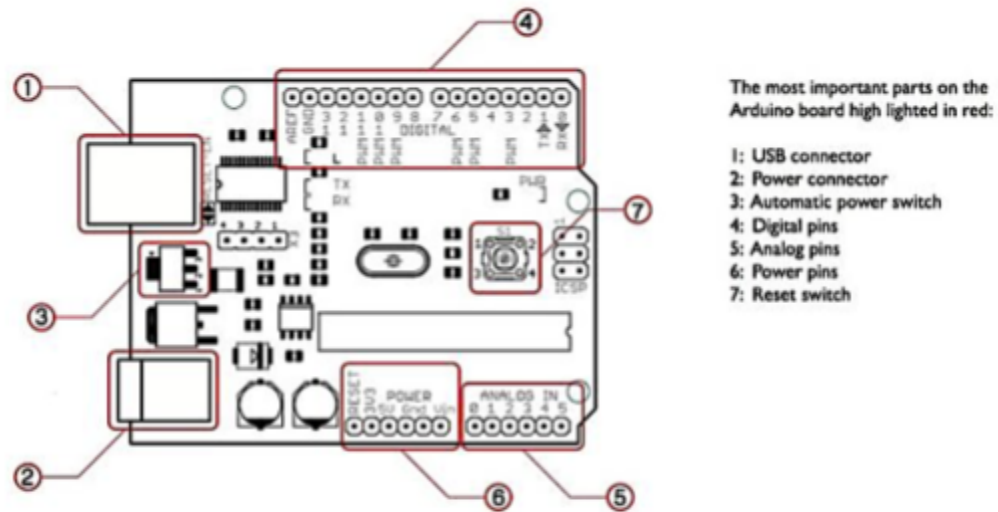
3.6 Architecture of Arduino



Figure[11]. Arduino Architecture

Arduino's processor basically uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories- Program memory and the data memory. The code is stored in the flash program memory, whereas the data is stored in the data memory. The Atmega328 has 32 KB of flash memory for

storing code (of which 0.5 KB is used for the bootloader), 2 KB of SRAM and 1 KB of EEPROM and operates with a clock speed of 16MHz.



Figure[12]. Arduino pin diagram

Arduino can be powered using either a USB connection to a computer or an external power source such as an adapter or battery. It can operate with an external voltage supply ranging from 7 to 12 volts. The external power source can be applied through pin Vin or by providing a voltage reference via the IOREf pin.

The digital inputs on Arduino consist of 14 pins, each capable of supplying or drawing up to 40mA of current. Several of these pins have specialized functions, such as pins 0 and 1 which act as the receiver (Rx) and transmitter (Tx) for serial communication, pins 2 and 3 which serve as external interrupts, and pins 3, 5, 6, 9, and 11 which provide pulse width modulation (PWM) output. Additionally, pin 13 is used to connect an LED.

The analog inputs on Arduino are comprised of 6 pins, each with a resolution of 10 bits. The analog reference (ARef) pin provides a reference voltage for the analog inputs. The reset pin will reset the microcontroller when it is held low.

3.7 Conclusion

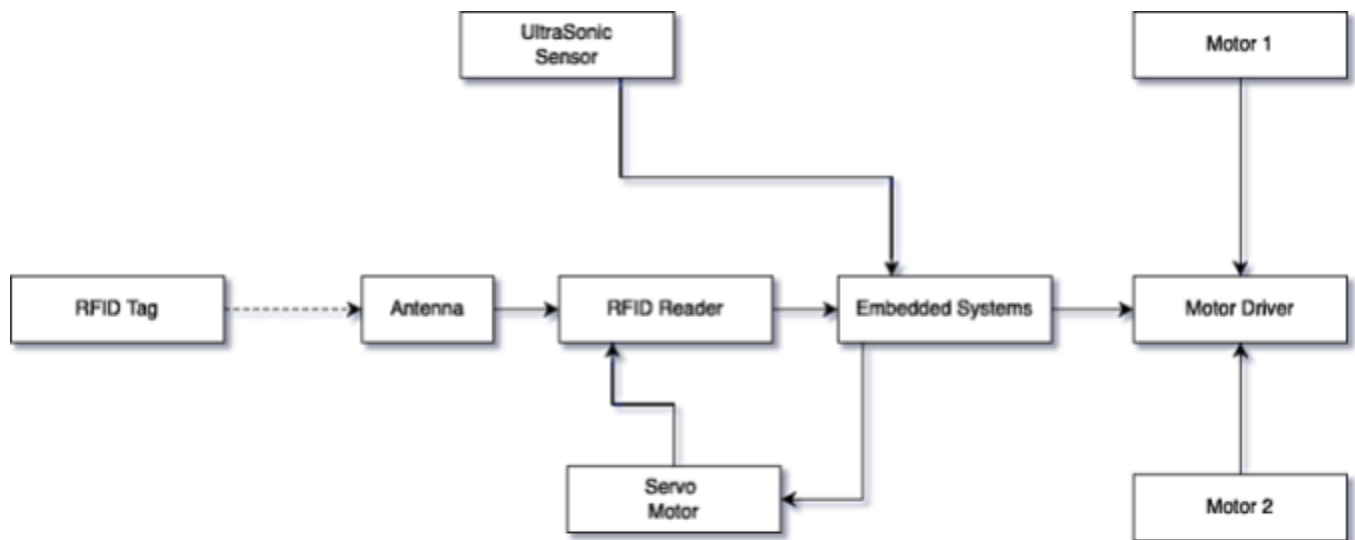
In summary, the integration of RFID technology with Arduino microcontroller platforms offers a compelling solution for building intelligent, connected systems with a wide range of applications. Whether it's tracking inventory in retail stores, monitoring environmental conditions in agricultural settings, or managing access control in smart buildings, the combination of RFID and Arduino technology empowers developers to create innovative solutions that enhance efficiency, transparency, and convenience in various domains.

SYSTEM DESIGN

4.1 Motivation

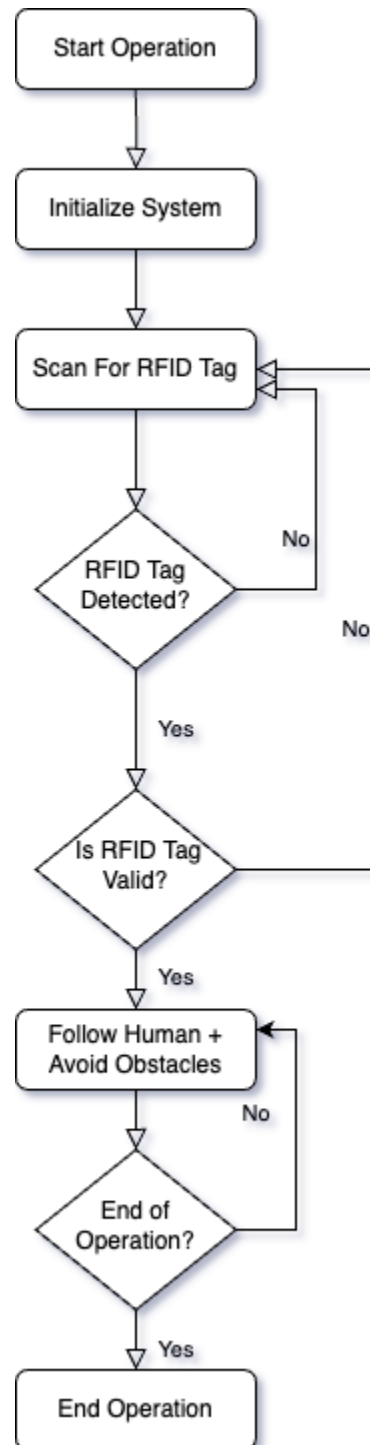
Simulations provide a controlled environment to test the efficiency, effectiveness, and safety of the robot's design before implementation in real-world scenarios. This allows for the identification of potential design flaws, enabling improvements and optimizations that lead to better overall performance.

4.2 Block Diagram



Figure[13]. Block diagram

4.3 Flow Chart



The process commences with the initiation of the robot's operation.

The robot's systems, including the RFID reader and ultrasonic sensors, are activated, and the environment is scanned for the presence of an RFID tag.

The system then checks for the detection of an RFID tag, and if one is present, it proceeds to verify its validity.

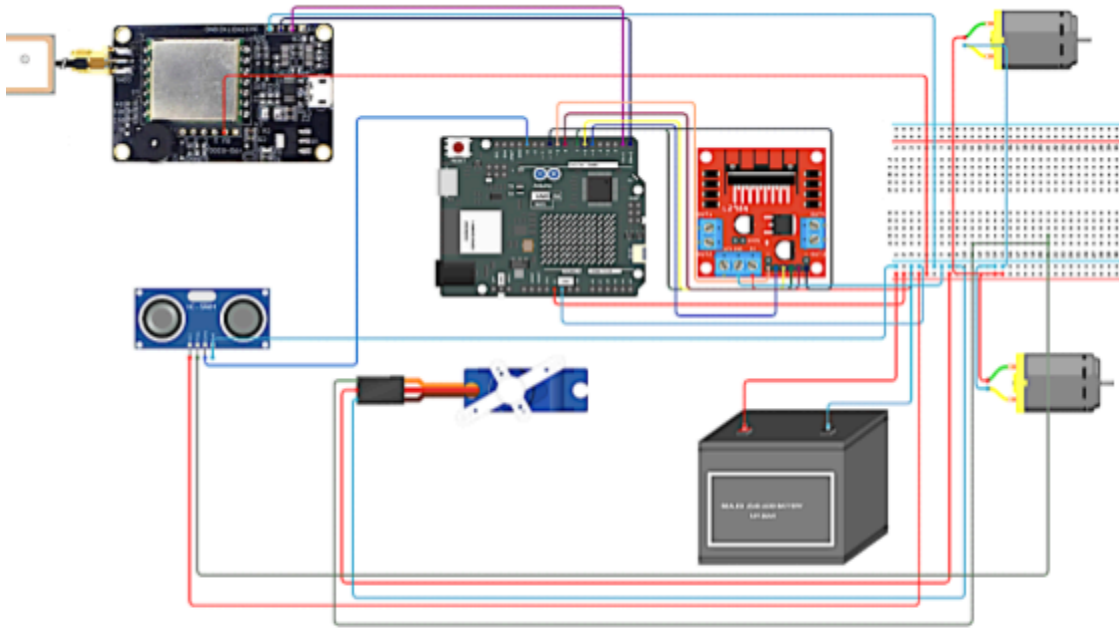
If the RFID tag is deemed invalid, the process returns to the scanning phase.

If the RFID tag is deemed valid, the robot follows the human carrying the tag while simultaneously employing its ultrasonic sensors to navigate around obstacles in its path.

The operation concludes once the robot has completed following the human or encounters any termination condition.

In essence, the flowchart outlines the steps the robot takes to function autonomously by detecting and adhering to a human carrying a specific RFID tag, whilst simultaneously avoiding obstructions in its path. The robot only follows valid RFID tags and continues scanning for them if an invalid tag is detected.

4.4 Initial Simulation



Figure[15]. Circuit diagram

4.5 Components

Table[1]. Components of follow me robot

No.	Component	Quantity
1	Arduino UNO	1
2	RFID Reader (IN-R200)	1
3	RFID Tag	1
4	2-wheel drive,Robot Chassis with Motors	1
5	Motor Driver (L298N)	1

6	Jumper Wires	As needed
7	Power Source (Battery Pack)	1
8	Ultrasonic Sensor	1
9	Breadboard	1
10	Circular Antenna	1

4.6 Circuit Connection

Table[2]. Arduino Uno Connections

Arduino Uno Pin	Connections
5v	BreadBoard
GND	BreadBoard
0(RX)	RFID Reader (IN-R200) TX
1(TX)	RFID Reader (IN-R200) RX
5,6,7, and 8	Motor Driver (L298N) IN1,IN2,IN3, and IN4
9 and 10	Motor Driver(L298N) ENA and ENB
12(PWM)	BreadBoard
13	Ultrasonic Sensor(HC-SR04) Echo

Table[3]. RFID Reader Connections

RFID Reader (IN-R200)	Connections
Vcc	Arduino Uno 5v
GND	Arduino Uno GND
TX	Arduino Uno 0(RX)
RX	Arduino Uno 1(TX)

Table[4]. Motor Driver (L298N) Connections	
Motor Driver (L298)	Connections
Vcc	Arduino Uno 5v
GND	Arduino Uno GND
IN1, IN2, IN3, and IN4	Arduino Uno pins 5,6,7, and 8
OUT 1	Left Motor Positive Wire

OUT 2	Left Motor Negative Wire
OUT 3	Right Motor Positive Wire
OUT 4	Right Motor Negative Wire

Table[5]. Servo Motor (SG90) Connections

Servo Motor (SG90)	Connections
Vcc (Red wire)	Arduino Uno 5v
GND (Black wire)	Arduino Uno GND
Signal (Orange wire)	Arduino Uno pin 12 (PWM)

Table[6]. Ultrasonic Sensor (HC-SR04) Connections

Ultrasonic Sensor	Connections
--------------------------	--------------------

(HC-SR04)	
Vcc	Arduino Uno 5v
GND	Arduino Uno GND
Trig	Arduino Uno pins 12
Echo	Arduino Uno pin 13

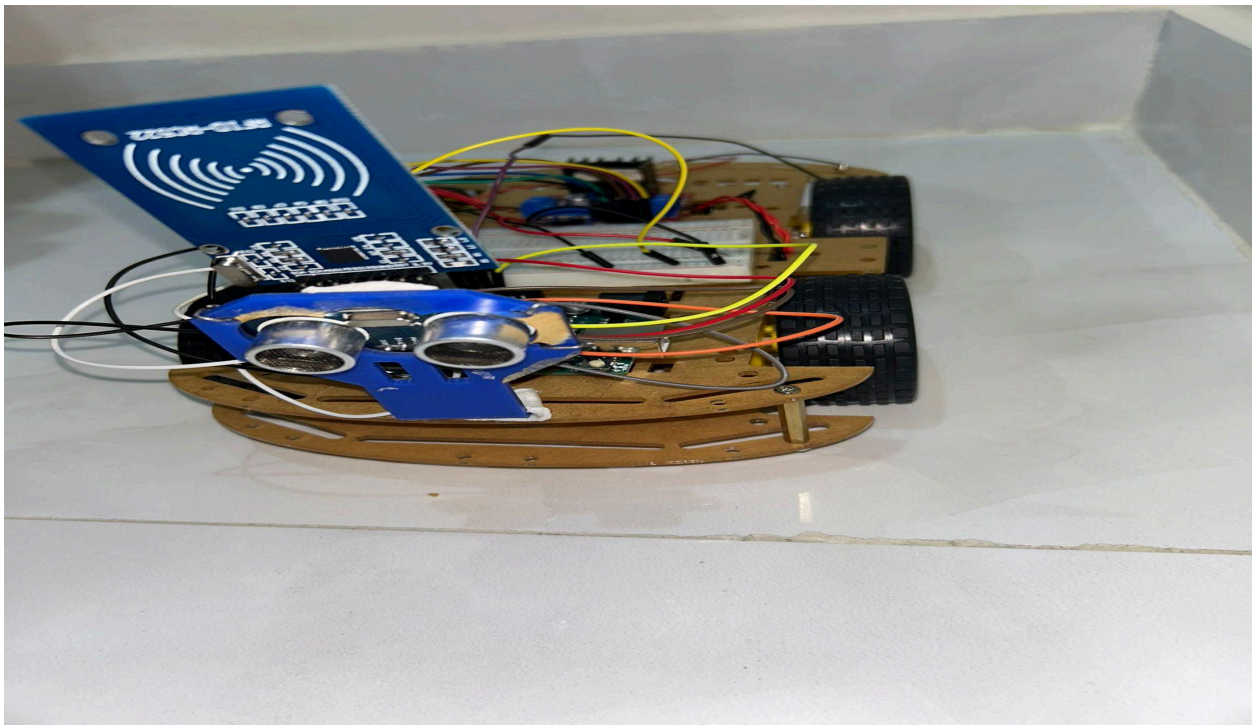
Table[7]. DC Motor Connections

DC Motor	Connections
Left Motor Positive Wire	Motor Driver OUT 1
Left Motor Negative Wire	Motor Driver OUT 2
Right Motor Positive Wire	Motor Driver OUT 3
Right Motor Negative Wire	Motor Driver OUT 4

IMPLEMENTATION

5.1 Introduction

The implementation phase is a critical part of the project lifecycle, transforming theoretical designs and conceptual plans into a functioning, real-world system. In this project, the goal was to build an autonomous human-following robot using RFID technology, integrating multiple hardware components and programming them to work seamlessly together.



5.2 Implementation Challenges and Adjustments

During the implementation phase, two significant challenges were encountered that necessitated adjustments to the initial design.

First, the UHF RFID R200 reader failed to retrieve the unique tag numbers reliably. Despite extensive troubleshooting, the issue persisted, likely due to compatibility limitations. To resolve this, the RFID R200 was replaced with the RFID RC522 module, which operates at 13.56 MHz and is well-supported by Arduino libraries. This change required adjustments to both the hardware configuration and the software code, but the RC522 proved to be reliable for tag detection and identification.

Second, the Arduino R4 microcontroller posed compatibility issues with the RFID library used for the RC522. This incompatibility prevented the system from functioning as intended. As a solution, the Arduino R4 was replaced with the Arduino R3, which ensured full library compatibility and seamless integration with the RC522 module.

These changes, though unforeseen, highlight the importance of flexibility in implementation. Addressing these challenges allowed the project to maintain progress and achieve the desired functionality.

5.3 Integration of subsystems

The integration of subsystems is a crucial step in ensuring the seamless functionality of the human-following robot. This phase involves combining individual hardware components and software modules to work cohesively as a unified system. In this project, the integration process focused on establishing effective communication between the RFID reader, ultrasonic sensors, motor driver, and the Arduino microcontroller to achieve real-time human tracking and obstacle avoidance.

- **Hardware Integration**

The hardware integration process began with connecting the RFID RC522 module, ultrasonic sensor, motor driver (L298N), and the robot's power supply to the Arduino R3 microcontroller. Each component was wired according to its specific pin configuration, ensuring compatibility and optimal signal transmission. Key challenges during this phase included managing power distribution across components and ensuring reliable data transfer between the RFID module and the microcontroller. These issues were mitigated by careful calibration and rigorous testing of the connections.

- **Software Integration**

The software integration involved writing and combining modular code for each subsystem. The RFID library was used to read tag data from the RC522 module, while ultrasonic sensor data was processed to detect obstacles. Motor control logic was implemented to drive the motors based on inputs from both the RFID and ultrasonic sensors. The Arduino platform facilitated the synchronization of these inputs and outputs, ensuring smooth communication between subsystems.

5.4 Software Development

The following code was written in the Arduino IDE and is responsible for the robot's primary functionalities. The RFID tag UID was updated to match the provided value: A9 EC F4 B8.

```
#include <SPI.h>
#include <MFRC522.h>

// RFID RC522 Pins
#define SS_PIN 10
#define RST_PIN 9
MFRC522 rfid(SS_PIN, RST_PIN);
```

```

// Ultrasonic Sensor Pins
#define TRIG_PIN A1
#define ECHO_PIN A2

// Motor Driver Pins
#define IN1 4
#define IN2 5
#define IN3 6
#define IN4 7

// Define RFID tag UID for following mode
byte validUID[] = {0xA9, 0xEC, 0xF4, 0xB8}; // Updated UID
bool tagDetected = false;

// Function prototypes
void moveForward();
void stopMotors();
bool isObstacleDetected();
bool isTagDetected();

void setup() {
  // Motor pins as outputs
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // Ultrasonic sensor pins
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  // RFID initialization
  SPI.begin();
  rfid.PCD_Init();

  Serial.begin(9600); // For debugging
  Serial.println("Robot Ready");
}

void loop() {
  tagDetected = isTagDetected();

  if (tagDetected) {
    if (isObstacleDetected()) {
      stopMotors(); // Stop if obstacle is detected
    }
  }
}

```

```

    Serial.println("Obstacle detected. Stopping.");
} else {
    moveForward(); // Move forward if tag is detected and no obstacles
    Serial.println("Tag detected. Moving forward.");
}
} else {
    stopMotors(); // Stop if tag is not detected
    Serial.println("No tag detected. Stopping.");
}

delay(100); // Loop delay for stability
}

// Function to check for obstacle using ultrasonic sensor
bool isObstacleDetected() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);
    int distance = duration * 0.034 / 2; // Distance in cm

    return distance < 20; // Obstacle detected if within 20 cm
}

// Function to detect RFID tag
bool isTagDetected() {
    if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
        return false; // No tag detected
    }

    // Compare detected UID with valid UID
    for (byte i = 0; i < rfid.uid.size; i++) {
        if (rfid.uid.uidByte[i] != validUID[i]) {
            rfid.PICC_HaltA(); // Halt communication if UID doesn't match
            return false;
        }
    }

    rfid.PICC_HaltA(); // Halt communication after successful read
    return true;
}

```

```
// Function to move forward  
void moveForward() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}  
  
// Function to stop motors  
void stopMotors() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, LOW);  
}
```

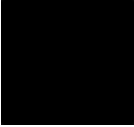
5.5 Future Improvement

The "Follow Me" robot presents a promising solution for RFID-based human tracking and obstacle avoidance, but several enhancements can further advance its functionality, efficiency, and versatility. One significant area of improvement is the sensing capability. Integrating advanced sensors like LiDAR could enhance the precision of distance measurements and obstacle detection, while incorporating a camera module with computer vision could enable more sophisticated tracking, such as facial recognition and dynamic path planning. Additionally, adding a GPS module would allow for outdoor navigation and precise localization, expanding the robot's range of applications.

CONCLUSION

6.1 Conclusion

In conclusion, the development of a mobile robot with RFID-based human following capabilities offers an innovative solution for various applications, such as cargo handling, elderly care, and more. The proposed robot integrates various components, including an Arduino Uno R4 WiFi as the central microcontroller, an IN-R200 RFID reader, a two-wheel drive robot chassis with L298N motor driver, and an HC-SR04 ultrasonic sensor for obstacle detection. By leveraging RFID technology, the robot can accurately track and follow a specific human target, ensuring efficient and safe operations in diverse environments. The integration of the ultrasonic sensor allows the robot to avoid obstacles during movement, enhancing its autonomous navigation capabilities. Furthermore, the use of a two-wheel drive chassis enables the robot to achieve precise movement control and rapid response to changes in the target's location. In summary, the combination of advanced technologies and innovative design makes this mobile robot a promising solution for addressing the challenges associated with human-robot interactions in various domains. Further research and development efforts can focus on optimizing its performance, extending battery life, and improving the robot's ability to adapt to different environments and applications.



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